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PI Name:	Farouk, Bakhtier Ph.D.		
Project Title:	Thermoacoustic Convection and Transport in Gases and Near-Critical Fluids under Microgravity Conditions		
Division Name:	Physical Sciences		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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PI Email:	bfarouk@coe.drexel.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	215-895-2287
Organization Name:	Drexel University		
PI Address 1:	Mechanical Engineering and Mechanics		
PI Address 2:	Main Building/Eng Lab 4-170D		
PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19104-2816	**Congressional District:**	2
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Contact Monitor:	Hall, Nancy Rabel	**Contact Phone:**	216-433-5643
Contact Email:	nancy.r.hall@nasa.gov		
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The research project is aimed at studying thermoacoustic and buoyancy-driven transport in gases and fluids near their critical points at normal and reduced gravity environments. It is envisioned that supercritical fluid reactors will play a vital role in the development of ‘water recovery’ and waste treatment systems in long duration manned space missions. The present four-year project (start date: December 8, 2003) combines computational and experimental studies of thermoacoustic waves and thermoacoustic convection in gases and near-critical fluids. In the first year of the project, significant advances have been made in both the model development and the experimental design and facility development. Large-scale detailed computational models were developed and tested to simulate, and understand the thermoacoustically driven flows and transport in gases and the generation and propagation of thermoacoustic waves in supercritical carbon dioxide. Both normal-gravity and reduced-gravity experiments are planned to validate the key features of the simulations. We extended our computational fluid dynamic models (for predicting thermoacoustic waves

Task Description:	in supercritical fluids) by incorporating the NIST Reference Database 12 for supercritical carbon dioxide properties. In the case of low-heat-diffusing, near-critical fluids, it has been recently shown that thermoacoustic waves, viz. the 'piston effect' is responsible for fast heat transport. The piston-effect has been successfully captured in our simulations using the models developed. Normal gravity experiments were conducted for the generation and characterization of thermoacoustic waves in gases. Preliminary data agree well with model predictions. An experimental setup to generate and characterize the thermoacoustic waves in supercritical carbon dioxide will be designed and fabricated. Studies will be carried out under both normal and reduced gravity environments. The present research is aimed at developing advanced computational tools to analyze interactions between thermoacoustic waves and viscous fluids in several problems of significant relevance to microgravity conditions. The specific objectives of the proposed research are the following: • Improve and refine a high-order computational model for the simulation of flow and heat transfer in gases and near-critical fluids induced by thermoacoustic waves • Perform direct numerical simulation of thermoacoustically driven gas flows in 2-D and 3-D enclosures • Completely characterize the behavior of the thermoacoustic waves (generation, propagation, interaction with solid walls and eventual dissipation) in gases and nar-critical fluids • Perform normal-gravity experiments for thermoacoustic waves in gases for model validation • Perform reduced-gravity experiments (short-time) in drop-towers or KC 135 aircraft for isolating buoyancy effects from thermoacoustic wave generated flows in gases • Select appropriate equation of state for near-critical fluids and incorporate the equation in the numerical model for flow simulation in such fluids • Perform direct numerical simulation of the piston effect in near-critical fluids in enclosures • Execute model for predicting the 'thermoacoustic' effects in near-critical fluids with and without gravitational effects • Perform normal-gravity experiments with near-critical CO2 in carefully designed experiments (which reduce gravitational effects) • Develop guidelines for the use of the mathematical models as a design and optimization tool for property and transport characterization in supercritical fluids and supercritical reactors. Successful completion of the research tasks will provide us important information regarding the design and operation of supercritical reactors. Such reactors are envisioned for waste management and water recovery in long duration manned space missions. The effects of thermoacoustic waves on transport and mixing in microgravity environments in such reactors willbe significant. The present research will be useful for the design of supercritical carbon dioxide reactors used in the food and pharmaceutical industries and for supercritical water reactors for waste management. The presence of gravitational field leads to both practical and fundamental limits of the resolution in critical phenomena experiments in fluids near the gas-liquid critical point. Near the gas-liquid critical point, gravity induces a density gradient in the fluid. The resulting inhomogeneous distribution of the local properties of the fluid strongly affects the interpretation of critical phenomena experiments in fluids. Near the critical point, fluids exhibit anomalies in the transport of heat such that convection and buoyancy phenomena appear for even minute temperature gradients. In the case of low-heat-diffusing, near-critical fluids, it has been shown recently that thermoacoustic waves, viz. the 'piston effect' is responsible for fast heat transport. A significant question regarding the 'piston effect' is whether the fast heat transport is due to the strong buoyant convection induced by normal gravity in near-critical fluids or if it is due to the high compressibility of such fluids. Microgravity environment is thus ideal for studying such processes for fluids near the critical point where new fluid physics can be discovered.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Supercritical fluids exhibit a number of specific properties like large density, low viscosity, low thermal diffusivity. In addition, their isothermal compressibility and thermal expansion can become very large, especially when they approach the critical point. The highly variable properties of near-critical fluids make them very attractive for studying many interesting phenomena that hold for all fluids because of the critical universality. Fluids in their supercritical state are increasingly used by the food and waste management industry for their solubilization properties (e.g. supercritical carbon dioxide), as host of "cold" combustion (e.g. supercritical water), in energetics (supercritical thermal or nuclear power plants) and in astronautics (e.g. storage of cryogenic fluids). It is particularly challenging to understand the behavior, flow and transport in a fluid that is close to its critical point. Near the gas-liquid critical point, gravity induces a density gradient in the fluid. The resulting inhomogeneous distribution of the local properties of the fluid strongly affects the interpretation of critical phenomena experiments in fluids. The research results will provide valuable insights to the trasnport processes in supercritical reactors - whether they are used on eart or long duration manned space missions.
	The research project is aimed at studying thermoacoustic and buoyancy-driven transport in gases and fluids near their critical points at normal and reduced gravity environments. It is envisioned that supercritical fluid reactors will play a vital role in the development of 'water recovery' and waste treatment systems in long duration manned space missions. Fluids in their supercritical state are increasingly used by the food and waste management industry for their solubilization properties, as host of "cold" combustion, in energetics and in astronautics. The present four-year project (start date: December 8, 2003) combines computational and experimental studies of thermoacoustic waves and thermoacoustic convection in gases and near-critical fluids. In the first year of the project, significant advances have been made in both the model development and the experimental design and facility development. Large-scale detailed computational models were developed and tested to simulate, and understand the thermoacoustically driven flows and transport in gases and the generation and propagation of thermoacoustic waves in supercritical carbon dioxide. The full Navier-Stokes equations for a compressible fluid are being considered for simulating the flow, pressure and temperature fields in an enclosure where thermoacoustic waves are produced by rapid

Task Progress:	heating. We have completed the design and fabrication of an experimental setup for characterizing thermoacoustic waves in gases. Sensitive microphones are used to measure the temporal variation of the acoustic waves at given locations. Normal and reduced gravity experiments are planned where careful measurements and visualization studies will be carried out to characterize thermoacoustic waves and heat transport in gases. Both normal-gravity and reduced-gravity experiments are planned to validate the key features of the simulations. For predicting thermoacoustic waves in supercritical fluids due to rapid heating, we carefully surveyed the available equations of state for near-critical carbon dioxide. The NIST Reference Database 12 was determined to be the most versatile and robust resource in this regard. We extended our computational fluid dynamic models (for predicting thermoacoustic waves in supercritical fluids) by incorporating the NIST database for supercritical carbon dioxide properties. In the case of low-heat-diffusing, near-critical fluids, it has been recently shown that thermoacoustic waves, viz. the 'piston effect' is responsible for fast heat transport. The piston-effect has been successfully captured in our simulations using the models developed. Normal gravity experiments were conducted for the generation and characterization of thermoacoustic waves in gases. Preliminary data agree well with model predictions. We plan to improve the experimental set-up next year such that ground-based reduced gravity experiments are possible in the 2.2 s zero gravity facility at the Glenn Research Center employing the NASA developed 'laser feedback interferometry'. An experimental setup to generate and characterize the thermoacoustic waves in supercritical carbon dioxide will be designed and fabricated. Studies will be carried out under both normal and reduced gravity environments. Near the gas-liquid critical point, gravity induces a density gradient in the fluid. Microgravity environment will be thus ideal for studying such processes for fluids near the critical point.
Bibliography Type:	Description: (Last Updated: 10/19/2009)
Papers from Meeting Proceedings	Lei Z, Farouk B, Aktas MK, Oran ES. "Generation and Propagation of Thermoacoustic Waves in Mildly Supercritical Carbon Dioxide." ASME International Mechanical Engineering Congress and Exposition, Anaheim, California, November 13- 19, 2004. Proceedings of the ASME International Mechanical Engineering Congress and Exposition, November 13- 19, 2004, Anaheim, California, Volume 1, 2004. , Nov-2004
Presentation	Farouk, B. "Thermoacoustically Induced Flows in Gases and Supercritical Fluids" Mechanical Engineering Department, Villanova University, Villanova, Pa. Mar-2004